CPME-II: Phase-Consistent Hebbian Consolidation

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CPME-II: Phase-Consistent Hebbian Consolidation

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A phase-coincidence extension to the ZeroGrad Hebbian rule that closes the chirality–consolidation gap identified in the CPME §6 failure modes. Chirality imprinted on injection now persists in long-term memory rather than dissipating over turns.

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0. Position

Chiral Phase-Momentum Emission (CPME, `research-paper/cpme.md`) encodes word order on the injection side as arrival-time phase and decodes structured emission on the readout side by sweeping a carrier phase across a per-cavity phase field $\phi_i = \text{atan2}(v_i, \omega_0 x_i)$. Both ends of the substrate now operate on phase.

The middle — long-term consolidation — does not. The ZeroGrad Hebbian rule inherited from the CMD substrate updates tunnel plasticity from the product of endpoint velocities:

$$\Delta w_{ij} \;\propto\; \eta \cdot v_i \cdot v_j$$

with a tanh-capped saturating integrator. This rule is phase-agnostic: it fires equally whether the two endpoints move in-phase, anti-phase, or in orthogonal phases. As a consequence, the chirality of the injection wavefront is not stored in the graph. Between turns the temporal ordering of concepts dissipates and must be re-imposed by each new input.

CPME-II adds a *phase-coincidence modulator* to the existing rule:

$$\Delta w_{ij} \;\propto\; \eta \cdot v_i \cdot v_j \cdot \cos(\theta_i - \theta_j)$$

where $\theta_i = \text{atan2}(v_i, \omega_0 x_i)$ is the same per-cavity phase the CPME emission layer already uses. The modulator preserves the velocity-based structure of the parent rule — including its sign, its tanh cap, and its role in the SRG dual-port Hebbian split — while conditioning the update on temporal compatibility.

The result is a substrate in which chirality is not only encoded and projected but also *stored*. Word order becomes a property of the graph, not just of the current turn.

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1. The Rule

1.1 Definition

For a tunnel connecting cavities a and b, define the instantaneous phase difference

$$\Delta\theta_{ab} = \theta_a - \theta_b \;=\; \text{atan2}(v_a, \omega_0 x_a) - \text{atan2}(v_b, \omega_0 x_b)$$

and the phase-coincidence factor

$$\Phi_{ab} = \cos(\Delta\theta_{ab})$$

taking values in $[-1, +1]$. The CPME-II Hebbian integrand replaces the parent rule's velocity product with a phase-modulated one:

$$I_{ab} \;=\; w_{ab} \cdot \zeta \;+\; \eta \cdot v_a^{\text{eff}} \cdot v_b^{\text{eff}} \cdot \Phi_{ab}$$

where ζ is the pre-existing Hebbian decay (`hebbian_decay`), η is the pre-existing rate scaled by serotonin (`plastic_rate`), v_a^{eff} and v_b^{eff} are the SRG-routed velocities already computed by the parent rule, and w_{ab} is the plasticity delta. The tanh saturation on `plasticity_cap` is applied unchanged.

The rule reduces to the parent rule exactly when $\Phi_{ab} = 1$ (perfect phase agreement) and vanishes when $\Phi_{ab} = 0$ (quadrature). For $\Phi_{ab} = -1$ (anti-phase), the update flips sign, which is the correct behaviour: anti-phase pairs are carrying opposing rather than reinforcing evidence and should depress rather than strengthen their coupling.

1.2 Interpretation

Φ_{ab} is a *continuous coincidence detector on the phase manifold*. It is the continuous-oscillator analog of spike-timing dependent plasticity (STDP): pairs whose oscillations align in the same phase-of-firing are consolidated; pairs whose oscillations consistently disagree are attenuated.

Because the phase field is chiral (§CPME 1.1), Φ_{ab} implicitly encodes temporal order. A concept driven earlier in the input has accumulated more phase than a concept driven later. Downstream cavities that phase-lock behind the driven cavity therefore experience a specific $\Delta\theta$ characteristic of *following the input in time*. Consolidating along this gradient stores the observation that the second cavity typically follows the first, without any explicit representation of "before" or "after".

1.3 What is stored

Under the parent rule, the tunnel between `insulin` and `glucose` grows in proportion to their velocity co-activity — regardless of which arrived first or how they moved relative to one another. Under CPME-II, the same tunnel grows only when the two cavities oscillate *in phase* — that is, when they consistently participate in the same wavefront with the same temporal role. If in one context `insulin` drives `glucose` (subject-then-object phase relationship) and in another context `glucose` drives `insulin` (reverse phase), the two updates cancel: $\cos(+\Delta\theta) + \cos(-\Delta\theta) = 2\cos(\Delta\theta)$ if the magnitude is preserved, but the sign of $\Delta\theta$ toggles the co-activity gradient. The graph accumulates a *phase-averaged co-activity* rather than a raw one.

The empirical effect is that stable directional patterns ("metformin → AMPK → glucose down") consolidate faster than context-dependent bidirectional patterns.

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2. Composition

CPME-II modifies only the Hebbian integrand. All other substrate mechanisms are untouched.

- **CMD Curie gate** (`heat_T` → tunnel force scaling) is independent of the Hebbian pass and continues to operate.
- **VGHT voltage gate** modulates tunnel force in the tick's spring-and-damping pass, not the Hebbian pass. CPME-II observes the resulting velocities and phases as inputs; VGHT is upstream.
- **SRG dual-port routing** (`gate_E`, `gate_I`) is preserved: v_a^{eff} in the CPME-II expression is the SRG-selected velocity from the parent rule, not the raw observable. The polarity-driven excitatory/inhibitory split remains intact.
- **CPME emission layer** (§1.6, phase-collapse projection) is unchanged. CPME-II operates during propagation and settle; the emission sweep reads whatever state the Hebbian pass has consolidated.
- **Plasticity cap** (`plasticity_cap`) remains the saturating bound; the tanh envelope contains the Φ_{ab}-modulated integrand identically.

The tick loop retains its no-if/else contract: $\cos(\Delta\theta)$ is a smooth scalar, evaluated once per tunnel per tick, and multiplied into the existing velocity product.

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3. Contract and Parameters

CPME-II adds no new state. All required quantities are already maintained by the parent substrate:

- x_a, v_a, x_b, v_b: cavity displacements and velocities.
- ω_0: the base cavity frequency `cpme_base_omega` shared with the emission layer.
- η, ζ, `plasticity_cap`: existing Hebbian parameters.

The only tunable is a *phase weight* $\alpha_\Phi \in [0, 1]$ that linearly interpolates between the parent rule and the fully phase-modulated rule:

$$\Phi_{ab}^{(\alpha)} \;=\; (1 - \alpha_\Phi) \;+\; \alpha_\Phi \cdot \cos(\Delta\theta_{ab})$$

- $\alpha_\Phi = 0$: recovers the parent rule exactly (backwards compatibility test path).
- $\alpha_\Phi = 1$: full phase-consistent consolidation.
- Default: $\alpha_\Phi = 0.6$, weighting phase agreement heavily while preserving a floor of unconditional Hebbian growth.

This is the only free parameter introduced by CPME-II. All defaults *on*.

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4. Expected Dynamics

Scenario A — repeated same-order exposure. Over a session in which the user types variants of "metformin lowers glucose" several times, `metformin` and `glucose` cavities are consistently driven with the same relative arrival delay. Their phase difference $\Delta\theta$ is stable and small, $\Phi \approx +1$, and the tunnel between them consolidates rapidly under the CPME-II rule. Between turns, the phase relationship is preserved because it is now stored in the plasticity delta, not just in the transient field.

Scenario B — bidirectional exposure. If in different turns the user types both "insulin controls glucose" and "glucose signals insulin", the pairwise $\Delta\theta$ toggles sign turn-to-turn. Under the parent rule, both directions add to plasticity. Under CPME-II, one direction has $\Phi > 0$ and the other $\Phi < 0$: the updates partially cancel. The tunnel still grows if the velocity product magnitude is large — the parent contribution $(1 - \alpha_\Phi)$ preserves an unconditional growth floor — but more slowly. This is the correct behaviour for a bidirectional relationship: the graph should not overcommit to either direction in isolation.

Scenario C — silent turn. If a turn produces no meaningful propagation and both cavities remain quiescent, v_a and v_b are near zero and the update is negligible regardless of Φ. No change from parent-rule behaviour.

Scenario D — anti-phase input. If the input is structured to produce anti-phase drive on a pair (e.g., a contrastive construction where the two cavities are being posited *against* each other), $\Phi \approx -1$ and the CPME-II rule *decreases* the plasticity delta. This is the physical realisation of what a biological network implements through inhibitory long-term depression: consistent anti-phase co-activation should weaken, not strengthen, the excitatory bond.

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5. Failure Modes

Three failure modes documented before deployment.

1. **Phase noise near quiescence.** For a cavity with small x and small v, the phase $\text{atan2}(v, \omega_0 x)$ is dominated by numerical noise. Φ then fluctuates randomly and injects noise into consolidation. Mitigation: the velocity product $v_a v_b$ already suppresses this — near quiescence the update magnitude is negligible regardless of Φ. No additional guard is required, but a diagnostic observer may report spurious Φ readings on cold cavities; this is expected and harmless.

2. **Polarity–phase interaction.** Inhibitory tunnels (`polarity < 0`) already invert the phase relationship they drive: an excitatory input to a produces an anti-phase response in b. Under CPME-II, this manifests as $\Phi < 0$ and the update would *depress* an inhibitory tunnel it should *strengthen*. Correction: multiply Φ by the tunnel polarity sign, $\Phi'_{ab} = \Phi_{ab} \cdot \text{sgn}(p_{ab})$, so that inhibitory tunnels consolidate on the anti-phase pattern they are supposed to enforce. This is applied by default in the implementation.

3. **Slow phase drift washout.** If the base cavity frequency ω_0 is very low, phase evolves slowly and inter-turn phase differences accumulate to $\pm\pi$ over long propagation windows, causing Φ to oscillate through zero and reverse sign spuriously. Mitigation: the same ω_0 chosen for the CPME emission layer (default $\omega_0 = 3.0$ rad/s) yields a full phase revolution over ≈ 2 seconds of sim-time, which is longer than any single propagation window at `dt = 0.02` and ≤ 200 ticks per turn. Under default parameters this failure mode does not

activate; it is worth flagging if ω_0 is later reduced.

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6. Biological Analog

CPME-II is the continuous-oscillator equivalent of spike-timing dependent plasticity (STDP). Where STDP consolidates a synapse when the presynaptic spike precedes the postsynaptic spike by a few milliseconds, CPME-II consolidates a tunnel when the two endpoint oscillators maintain a stable, small phase offset consistent with the same wavefront driving both. Both rules use temporal coincidence as the substrate of long-term memory; both rules invert when the coincidence is anti-correlated (STDP-LTD; CPME-II negative Φ).

The chief difference is discretisation. STDP fires on discrete spike pairs; CPME-II integrates over the continuous phase field each tick. In the limit of narrow, well-separated velocity peaks, CPME-II reduces to a spike-coincidence rule up to a normalisation constant.

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7. Summary

CPME-II closes the chirality–Hebbian mismatch flagged in HTR §5.2 and CPME §6.2 by extending the ZeroGrad Hebbian rule with a phase coincidence factor $\Phi_{ab} = \cos(\theta_a - \theta_b)$. The factor requires no new state — the same per-cavity phase $\text{atan2}(v_i, \omega_0 x_i)$ used by the CPME emission layer is read during the Hebbian pass. A single blend parameter α_Φ controls how strongly the modulator gates the parent rule; the default $\alpha_\Phi = 0.6$ preserves an unconditional Hebbian floor while making chirality-consistent updates twice as strong as chirality-neutral ones on average.

The consequence for the substrate as a whole: the phase manifold that CPME already uses on the input (chirality) and the output (phase-collapse projection) is now also the manifold on which long-term memory is written. Word order becomes a graph property, temporal ordering persists across turns, and the substrate is freed from having to re-derive chirality on every input.